BOTANY HUB MULTAI

B.Sc. I BOTANY — MINOR I

ELEMENTARY BOTANY

UNIT V — Reproduction, Growth and Propagation

DETAILED TEACHER-STYLE NOTES

Definitions • Every concept explained • Tables • Labelled study diagrams • Flowcharts • Examples • Comparisons • Exam points

Prepared in the same detailed format as Major Course II and Major Course III notes.

UNIT V — REPRODUCTION, GROWTH AND PROPAGATION

The supplied notes cover sexual and asexual reproduction, vegetative propagation, plant growth regulators, gametophyte–sporophyte alternation and propagation methods. ■filecite■turn7file0■L140-L158■

1. Reproduction

Definition — Reproduction: The biological process by which organisms produce new individuals and maintain continuity of their lineage.

Feature	Asexual reproduction	Sexual reproduction
Gamete fusion	Absent	Present
Genetic outcome	Usually closely similar to parent, though mutations can occur	Produces new combinations through meiosis and fertilization
Speed	Often rapid	Usually more steps involved
Examples	Vegetative propagation, spores in suitable organisms	Seed formation in flowering plants

2. Vegetative propagation

Definition — Vegetative propagation: Formation of new plants from vegetative parts such as stem, root or leaf rather than from seed.

Method	Principle	Example
Cutting	A detached plant piece develops roots/shoots	Rose, sugarcane in suitable contexts
Layering	Roots are induced while branch remains attached initially	Jasmine
Grafting	Scion is joined to a compatible rooted stock	Mango
Division	Existing plant clump/structure is separated	Many ornamentals
Natural propagation	Modified organs produce new plants	Potato, ginger

3. Plant growth regulators

Definition — Plant growth regulators: Naturally occurring or synthetic compounds that, at low concentrations, influence plant growth, development and physiological responses.

Regulator	Major textbook role	Common study point
Auxin	Cell elongation and developmental responses	Apical dominance and rooting contexts
Gibberellin	Stem elongation and developmental regulation	Growth promotion in specific contexts
Cytokinin	Cell division and shoot-related responses	Tissue-culture relevance
Abscisic acid (ABA)	Stress and growth-restraint responses	Stomatal and dormancy-related contexts
Ethylene	Gaseous regulator with ripening and developmental effects	Fruit-ripening-related responses

4. Gametophyte and sporophyte

Definition — Gametophyte: The haploid (n) generation that produces gametes by mitosis.

Definition — Sporophyte: The diploid (2n) generation that produces spores by meiosis.

Alternation of generations

Sporophyte (2n) → Meiosis → Spores (n) → Gametophyte (n) → Gametes → Fertilization → new sporophyte

5. Growth and propagation strategy

Plant growth results from cell division, enlargement and differentiation. Propagation methods are selected according to the desired outcome: seed propagation is useful for genetic diversity and breeding, while vegetative propagation is preferred when maintaining a selected genotype is important.

Choice	Best when	Trade-off
Seed propagation	Variation or large-scale seed production is acceptable	Offspring may vary
Vegetative propagation	Uniform clones of selected plants are required	Disease can spread if source material is infected
Grafting	Desired shoot traits and suitable rootstock traits are combined	Requires compatibility and skill
Tissue culture	Rapid controlled multiplication is required	Needs sterile facilities and technical management

Exam focus: Unit V long answer format: definition → classification → mechanism/steps → comparison table → examples → importance. The supplied notes also recommend this structured approach rather than one long paragraph. ■filecite■turn7file0■L154-L166■